

Data Structure and Algorithms project

Title : Library management System.

Description :

The program will allow the **employee** to **add** the books and the sections into the system, **remove** a book and a section, **update** a book and a section, **sort** the books, **search** a book and a section, **display** books and sections, and see the **total** amount of books and sections.

Employee will have many options to do, which are:

- **Adding** a new section and new book (with its details).
 - **Details** are: Title, code, price, quantity, author, section.
 - **Removing** a section and a book.
 - **Updating** section or book details.
 - **Sorting** The books order.
 - **Searching** books, and sections.
 - **Seeing**, sections, and books details (display).
 - **Total** amount of books, sections.
-

Output format (Recommended) :

1- Employee/Main Menu:

```
|
|                                     Main Menu
|-----|
| 1 - Add a new ( book / section ).
| 2 - Remove a ( book / section ).
| 3 - Update a ( book / section ).
| 4 - Sort the books.
| 5 - Search a ( book / section ).
| 6 - display ( books / sections ).
| 7 - Total ( books / sections ).
| 8 - Exit.
|
|-----|
> Enter Your Choice :
```

2- Employee -> Add Menu:

```
=====
|                               Main -> Add Menu
|-----
| 1 - Add a new book.
| 2 - Add a new section.
| 3 - Back to the Main Menu.
|
|-----
> Enter Your Choice :
```

3- **Emplyee** → Add Menu → Add a book:

Enter The book number [] details.

> Title :
> Author :
> Code :
> Section (1- Fiction , 2- Acadimic , 3- Business) :
> Price (\$) :
> Availible Quantity :

Do you want to add another book (yes / no) ?

4- **Emplyee** → Add Menu → Add a section:

Sections you have:

1- Fiction.
2- Acadimic.
3- Business.

Do you want to add another section (yes / no) ?

//if(yes) - Write the new section name:

5- Employee → Remove Menu:

```
=====
|                               Main -> Remove Menu                               |
|-----|
| 1 - Remove book. |
| 2 - Add a new section. |
| 3 - Back to the Main Menu. |
| |
|-----|
> Enter Your Choice :
```

6- **Emplyee** → Remove Menu → Remove a book:

Books you have:

No.	Title	Author	Code	Section	Price	Quantity
1						
2						
3						
...						

> Enter the book [Code] to remove :

Do you want to remove another book (yes / no) ?

7- **Emplyee** → Remove Menu → Remove a section:

Sections you have:

1- Fiction.
2- Acadimic.
3- Business.
...

> Enter section's [number] to remove :

Do you want to remove another section (yes / no) ?

8- Emplyee -> Update Menu:

```
=====
|                               Main -> Update Menu                               |
|-----|
| 1 - Update a book.                                                     |
| 2 - Update a section.                                                 |
| 3 - Back to the Main Menu.                                           |
|-----|
> Enter Your Choice :
```

9- Emplyee -> Update Menu -> Update a book:

Books that library has:

```
=====
|No. |Title           |Author |Code  |Section |Price |Quantity |
|-----|
|1   |                |       |      |        |      |          |
|2   |                |       |      |        |      |          |
|3   |                |       |      |        |      |          |
|... |                |       |      |        |      |          |
=====
```

> Enter the book [Code] you want to Update :

Which one you want to update?

- 1- Title.
- 2- Author.
- 3- Section.
- 4- Price.
- 5- Availible Quantity.
- 6- All of them.

> Enter Your Choice :
- Write The new ... :

Do you want to update another book (yes / no) ?

10- **Employee** -> Update Menu -> Update a section:

Sections that library has:

```
=====
|No. |Section          |
-----
|1   |Fiction           |
|2   |Academic           |
|3   |Business           |
|... |                   |
=====
```

> Enter section's [number] you want to Update :
- Write the new Name :

Do you want to update another section (yes / no) ?

11- **Employee** -> Sort Menu:

```
=====
|                               Main -> Sort Menu                               |
-----
| 1 - Sort by [Title].           |
| 2 - Search by [Code].         |
| 3 - Sort by [Author].         |
| 4 - Sort by [Section].         |
| 5 - Sort by [Price].           |
| 6 - Sort by [Quantity].        |
| 7 - Back to the Main Menu.     |
|                               |
=====
```

> Enter Your Choice :

12- **Emplyee** → Search Menu:

```
=====
|                               Main -> Search Menu                               |
|-----|
| 1 - Search a book.                                                     |
| 2 - Search a section.                                                 |
| 3 - Back to the Main Menu.                                           |
|-----|
> Enter Your Choice :
```

13- **Emplyee** → Search Menu → Search a book:

Which way do you want to search by?

- 1 - Search by [Title].
- 2 - Search by [Code].
- 3 - Search by [Author].
- 4 - Search by [Section].
- 5 - Search by [Price].
- 6 - Back to the Main Menu.

> Enter Your Choice :

14- **Emplyee** → Display Menu:

```
=====
|                               Main -> Display Menu                               |
|-----|
| 1 - Display books.              |
| 2 - Display sections.          |
| 3 - Back to the Main Menu.     |
|                                |
|-----|
> Enter Your Choice :
```

15- **Emplyee** → Display Menu → Display books:

Books that library has:

```
=====
|No. |Title           |Author |Code |Section |Price |Quantity |
|-----|
|1   |                |       |     |         |      |          |
|2   |                |       |     |         |      |          |
|3   |                |       |     |         |      |          |
|... |                |       |     |         |      |          |
|-----|
```

16- **Employee** → Display Menu → Display sections:

Sections that library has:

=====		
No.	Section	

1	Fiction	
2	Academic	
3	Business	
...		
=====		

17- **Employee** → Total Menu:

=====		
	Main -> Total Menu	

1 -	Total amount of books.	
2 -	Total amount of sections.	
3 -	Back to the Main Menu.	
=====		
> Enter Your Choice :		

Functions / Tasks distribution :

1- **Name:** Reda Alqatifi.

- **Adding** (for employee).
- **Removing** (for employee).
- **Sorting** (for employee).
- **General system core.**
- **Merging everything together.**

2- **Name:** Mohammed Aljasem.

- **Searching** (for employee).
- **Presentation and report (less than 30 pages).**

3- **Name:** Mahdi Alhashim.

- **Updating** (for employee).
- **Presentation and report (less than 30 pages).**

4- **Name:** Ali Alburahim.

- **Displaying** (for employee).
 - **Total** (for employee).
-

*** Notes:**

- The output is **very recommended** to be like the one above, but if you couldn't do it, no problem do it in your way.
 - Please, use **clear names** for functions.
 - Please, use **comments** to make the code as clear as possible.
 - Everyone **must** finish from their Functions before this date : **2025/12/12** , Because we need some time to merge them together, test and debug them, and make the report.
 - Data Structures **mostly** we will use:
 - Doubly Linked List.
 - Linked List.
 - Merge sort.
-